Seat No: 05

Name:Nithish Vaibhav E

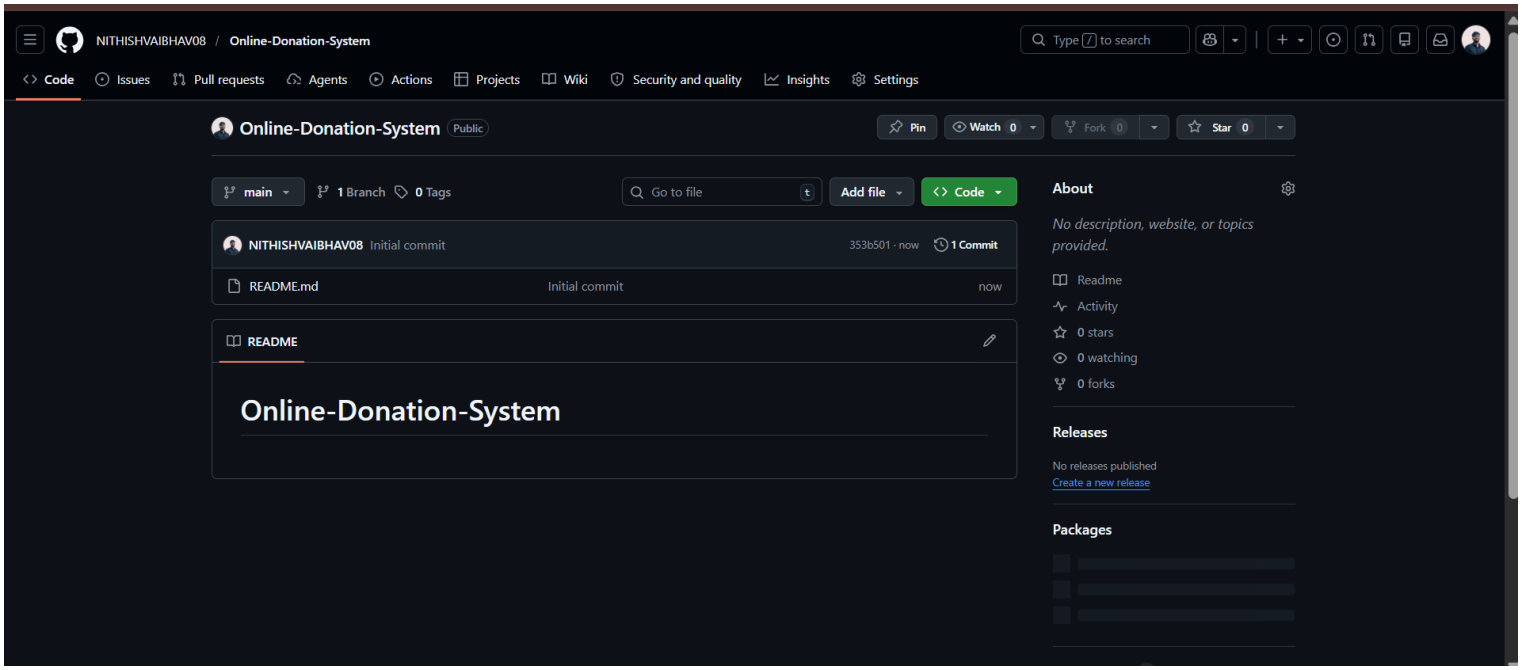
Reg no: 23MIS0680

Problem: Online Donation System

1.File Structure

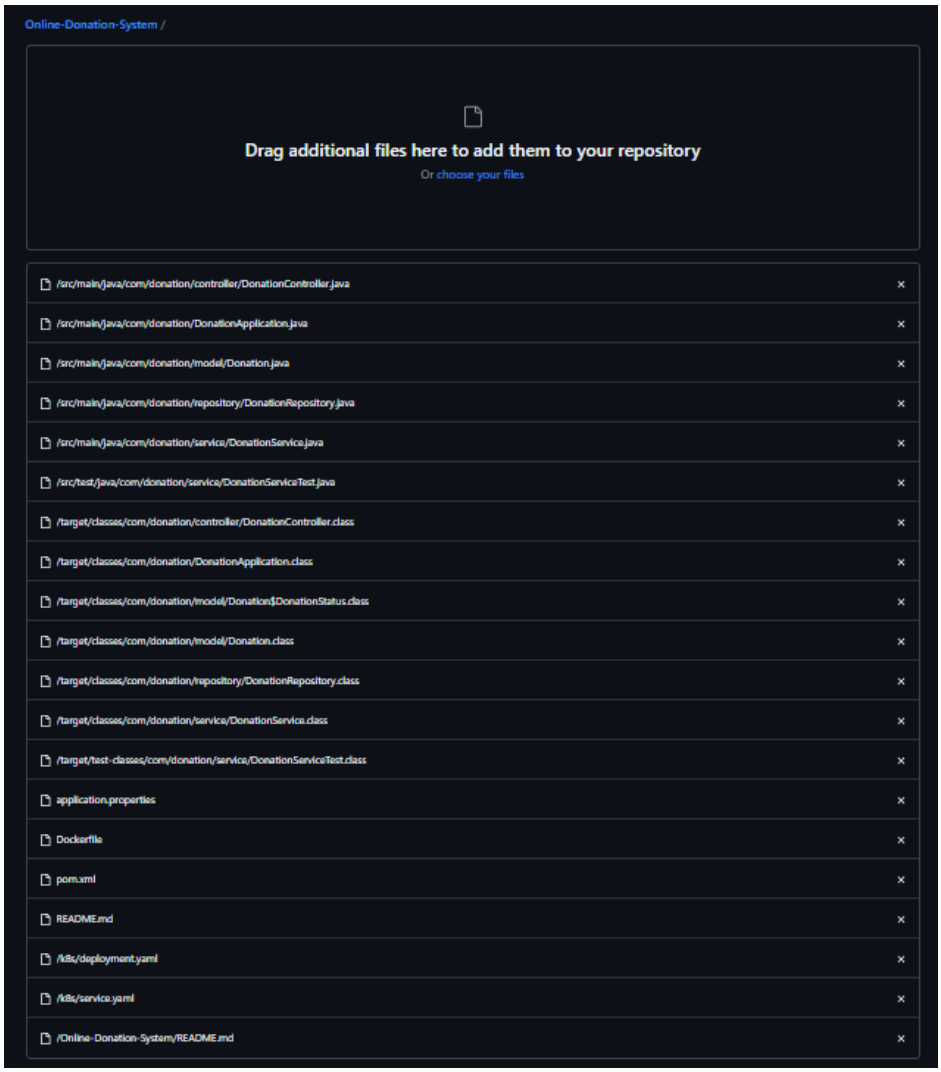
```
donation-system/
├── src/
│   ├── main/java/com/donation/
│   │   ├── DonationApplication.java      ← Spring Boot entry point
│   │   ├── model/Donation.java           ← Entity
│   │   ├── repository/DonationRepository.java
│   │   ├── service/DonationService.java  ← Business logic
│   │   └── controller/DonationController.java
│   └── test/java/com/donation/service/
│       └── DonationServiceTest.java      ← JUnit 5 tests (25 tests)
├── .github/workflows/ci-cd.yml           ← GitHub Actions pipeline
├── Dockerfile                            ← Multi-stage Docker build
├── jenkinsfile                           ← CI/CD Pipeline Code
├── k8s/
│   ├── deployment.yaml                   ← Kubernetes Deployment
│   └── service.yaml                      ← Kubernetes Service + Namespaces
└── pom.xml
```

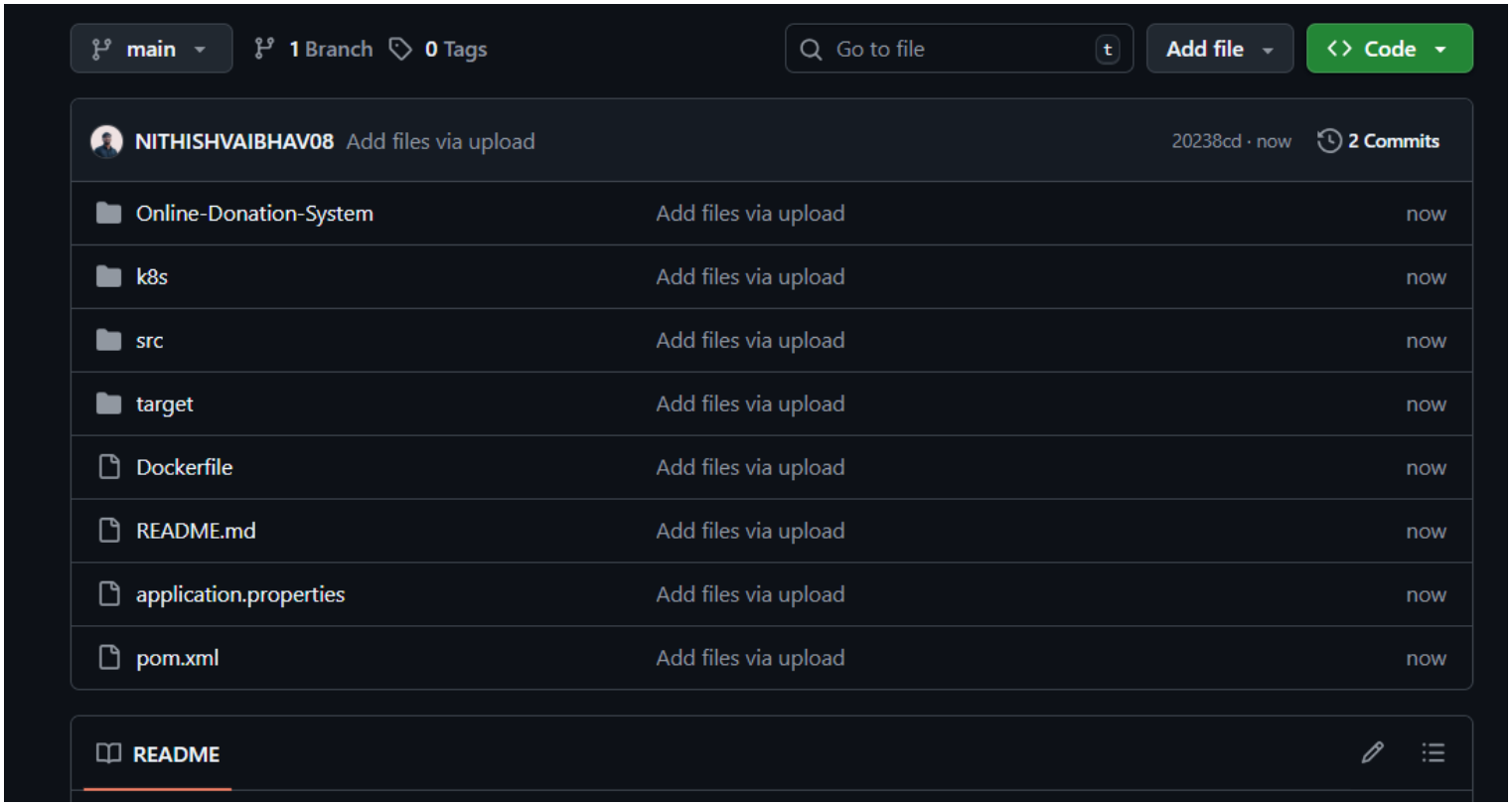
2.Create Git Repository for the java application:



3.Setting up version control using GitHub Uploading files from PC to repository and commiting changes:

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> git clone https://github.com/NITHISHVAIBHAV08/Online-Donation-System
Cloning into 'Online-Donation-System'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```





3.Run the Application Locally:

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system>
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.donation:online-donation-system >-----------------
[INFO] Building Online Donation System 1.0.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.3.2/maven-clean-plugin-3
.3.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.3.2/maven-clean-plugin-3.
3.2.pom (5.3 kB at 5.8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/40/maven-plugins-40.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/40/maven-plugins-40.pom (8.1 kB
at 44 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/40/maven-parent-40.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/40/maven-parent-40.pom (49 kB at 154 kB/s
)
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ online-donation-system ---
[INFO] Deleting C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ online-donation-system ---
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ online-donation-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 5 source files with javac [debug release 17] to target\classes
```

```
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ online-donation-system ---
[INFO] Deleting C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ online-donation-system ---
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ online-donation-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 5 source files with javac [debug release 17] to target\classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.419 s
[INFO] Finished at: 2026-04-15T18:13:35+05:30
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> mvn spring-boot:run
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/io/github/git-commit-id/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/jooq/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/flywaydb/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/jooq/maven-metadata.xml (398 B at 424 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/maven-metadata.xml (249 B at 265 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/flywaydb/maven-metadata.xml (234 B at 248 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/io/github/git-commit-id/maven-metadata.xml (255 B at 270 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/liquibase/maven-metadata.xml
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system>
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.donation:online-donation-system >-----------------
[INFO] Building Online Donation System 1.0.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ online-donation-system ---
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ online-donation-system ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ online-donation-system ---
```

```
[INFO] -------------------------------------------------------
[INFO] Running com.donation.service.DonationServiceTest
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been app
ended
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.328 s -- in com.donation.service.DonationServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.090 s
[INFO] Finished at: 2026-04-15T18:16:52+05:30
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.donation:online-donation-system >-----------------
[INFO] Building Online Donation System 1.0.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ online-donation-system ---
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ online-donation-system ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ online-donation-system ---
[INFO] skip non existing resourceDirectory C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\src\test\resources
[INFO]
```

```
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been app
ended
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.338 s -- in com.donation.service.DonationServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ online-donation-system ---
[INFO] Building jar: C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\target\online-donation-system-1.0.0.jar
[INFO]
[INFO] --- spring-boot:3.2.0:repackage (repackage) @ online-donation-system ---
[INFO] Replacing main artifact C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\target\online-donation-system-1.0.0.jar wit
h repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system\target\online-donation-sy
stem-1.0.0.jar.original
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.316 s
[INFO] Finished at: 2026-04-15T18:18:39+05:30
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system>
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> docker build -t auth-app:latest .
[+] Building 28.0s (10/18)                                                    docker:desktop-linux
 => [internal] load metadata for docker.io/library/maven:3.9.6-eclipse-temurin-17                7.9s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                           0.0s
 => [auth] library/maven:pull token for registry-1.docker.io                                     0.0s
 => [internal] load .dockerignore                                                                 0.0s
 => => transferring context: 2B                                                                   0.0s
 => [builder 1/6] FROM docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7  20.0s
 => => resolve docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7b519b45eb  0.0s
 => => sha256:fc06d68d71ba375a9ceed165c0c04b09b1a8cc193146888663b005660fc25013 156B / 156B        0.7s
 => => sha256:4712dfa85971124b8c3507fbba4c02441e5754f1517162926e600afd3fc5404b 358B / 358B        1.1s
 => => sha256:58c3491a14ebc973960cbda9f2eea22537785a3cb07e81487bd1115bad4a8278 852B / 852B        1.5s
 => => sha256:45f480637770dbb740623c0c7f827d44bbacc9204faf3d07b4135d54a2bee043 9.48MB / 9.48MB    8.3s
 => => sha256:67f99c2668af1d0eced6d61dad057d7beffe543a8e8370d5c0d3fe8682e059a3 19.00MB / 19.00MB  12.7s
 => => sha256:a3ba11f7aaaedad962216812fe84aee9061aeabac6932f0274a1d204eb96e8e8 734B / 734B        0.8s
 => => sha256:c97285723537ab7921fca4d081c256e501adfdaa8992d04637992075f4cea392 0B / 173B         18.3s
 => => sha256:d59fd278c1b4ffe1727be6ccba42125e8f4db57e660e795cce19889d2c776457 9.44MB / 145.10MB  17.9s
 => => sha256:5e5d1bccc5440d3a24f4a620704b9e687b4163c6c872fcc8e812e200c9bbac58 7.34MB / 17.46MB   17.4s
 => => sha256:4a023cab5400feb5c1ab725beb8345ddb0e3200314004b56677a5eee2e8c86cf 16.78MB / 30.44MB  11.5s
 => [internal] load build context                                                                 0.1s
 => => transferring context: 32.08kB                                                              0.0s
 => [stage-1 1/5] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:7aa804a1824d18d06c68598fe1  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:7aa804a1824d18d06c68598fe1c2953b5b  0.0s
```

```
 => => resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:7aa804a1824d18d06c68598fe1c2953b5b  0.0s
 => CACHED [stage-1 2/5] WORKDIR /app                                                                  0.0s
 => [stage-1 3/5] RUN addgroup -S donation && adduser -S donation -G donation                          0.4s
 => [builder 2/6] WORKDIR /app                                                                         0.3s
 => [builder 3/6] COPY pom.xml .                                                                       0.1s
 => [builder 4/6] RUN mvn dependency:go-offline -B                                                   161.9s
 => [builder 5/6] COPY src ./src                                                                       0.1s
 => [builder 6/6] RUN mvn clean package -DskipTests -B                                                12.6s
 => [stage-1 4/5] COPY --from=builder /app/target/online-donation-system-1.0.0.jar app.jar            0.1s
 => [stage-1 5/5] RUN chown donation:donation app.jar                                                  0.3s
 => exporting to image                                                                                 1.6s
 => => exporting layers                                                                                1.0s
 => => exporting manifest sha256:d6b1252eaabead565fbf2bafef062ac196c530eb63e95237d26341640d2c4b17      0.0s
 => => exporting config sha256:f864000d10fc6d1cef67e399069298d88fe259614d69ad9dedf842c31d63b699        0.0s
 => => exporting attestation manifest sha256:d591e24995bec05ab4836d7102b74121b163060fa523ecae0851e5ae416  0.0s
 => => exporting manifest list sha256:2bfb9e588e1724bf00e6ad2636ec345b220efcebd124101b37508b6764716573  0.0s
 => => naming to docker.io/library/auth-app:latest                                                     0.0s
 => => unpacking to docker.io/library/auth-app:latest                                                  0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/q8okuug2c391v03pp24km65ke
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> kubectl apply -f
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> kubectl apply -f k8s/service.yaml
service/donation-system-service created
namespace/staging unchanged
namespace/production unchanged
secret/donation-db-secret configured
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system>
```

```
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system> kubectl apply -f k8s/deployment.yaml
deployment.apps/donation-system created
PS C:\Users\Nithish vaibhav\Desktop\AGILE FAT\donation-system>
```

Deployment and build jar files are completed push it to git and start CI/CD Pipeline in Jenkins.

New Item

Enter an item name

Online Donation System

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

Copy from

Type to autocomplete

OK

GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

Note

Agile Lab Fat

What's this token for?

Expiration

7 days (Apr 22, 2026)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. Read more about OAuth scopes.

☑ **repo**	Full control of private repositories
☑ repo:status	Access commit status
☑ repo_deployment	Access deployment status
☑ public_repo	Access public repositories
☑ repo:invite	Access repository invitations
☑ security_events	Read and write security events
☑ **workflow**	Update GitHub Action workflows
☐ **write:packages**	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ **delete:packages**	Delete packages from GitHub Package Registry

GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Tokens (classic)

Personal access tokens (classic)

Generate new token

Tokens you have generated that can be used to access the GitHub API.

Make sure to copy your personal access token now. You won't be able to see it again!

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 Delete

nithish — *repo, workflow* Never used Delete

Expires ***on Tue, May 12 2026.***

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

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Jenkins / Manage Jenkins / Credentials / System / Global / nithishvaibhav08/****** (G...

Update
Delete
Move

Update credentials

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username

nithishvaibhav08

Treat username as secret ?

Password

••

ID ?

github-creds

Description ?

GitHub Access

Save

Jenkins / Online Donation System / Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition: Pipeline script from SCM

SCM: Git

Repositories

Repository URL: https://github.com/NITHISHVAIBHAV08/Online-Donation-System

Credentials: nithishvaibhav08/****** (GitHub Access) + Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any'): */main

+ Add Branch

Repository browser: (Auto)

Additional Behaviours

+ Add

Script Path: Jenkinsfile

☑ Lightweight checkout

Pipeline Syntax

Advanced

Advanced

Save | Apply

Jenkins / Online Donation System / #13

- Status
- Changes
- Console Output
- Edit Build Information
- Timings
- Git Build Data
- Pipeline Overview
- Thread Dump
- Pause/resume
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

#13 (Apr 15, 2026, 6:40:37 PM)

Progress: | Add description | Keep this build forever

Started by user Nithish Vaibhav E

Started 41 sec ago
Build has been executing for 41 sec

This run spent 5 ms waiting in the queue.

git
Revision: b0f97865e69a5b3d0ac2f90921eabb5f4a71a645
Repository: https://github.com/NITHISHVAIBHAV08/EXAM-EVALUATION-SYSTEM

- origin/main

No changes.

Search

Name	Tag	Image ID	Created	Size	Actions
gcr.io/k8s-minikube/k	<none>	eb4fec00e8ad	2 months ago	1.93 GB	
kindest/node	v1.34.3	08497ee19eac	4 months ago	1.35 GB	
docker/desktop-cloud	v0.5.0	4ad59ce20658	4 months ago	595.06 MB	
envoyproxy/envoy	v1.36.4	08acc53db6f4	4 months ago	247.75 MB	
docker/desktop-conta	v0.0.3	179a9ea3ae47	4 months ago	50.32 MB	
auth-app	latest	2bfb9e588e17	17 minutes ag	436.71 MB	
exam-app	latest	f2a102b55f74	1 minute ago	510.06 MB	

Showing 12 items

Java Code:

```
package com.donation;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class DonationApplication {
   public static void main(String[] args) {
      SpringApplication.run(DonationApplication.class, args);
   }
}
```

Java test Code:

```
package com.donation.service;

import com.donation.model.Donation;
import com.donation.repository.DonationRepository;
import org.junit.jupiter.api.*;
import org.junit.jupiter.api.extension.ExtendWith;
import org.mockito.*;
import org.mockito.junit.jupiter.MockitoExtension;

import java.math.BigDecimal;
import java.time.LocalDateTime;
import java.util.Arrays;
import java.util.List;
import java.util.Optional;

import static org.junit.jupiter.api.Assertions.*;
import static org.mockito.ArgumentMatchers.*;
import static org.mockito.Mockito.*;

@ExtendWith(MockitoExtension.class)
@DisplayName("DonationService Unit Tests")
```

```
class DonationServiceTest {

    @Mock
    private DonationRepository donationRepository;

    @InjectMocks
    private DonationService donationService;

    private Donation validDonation;

    @BeforeEach
    void setUp() {
        validDonation = new Donation(
                "John Doe",
                "john.doe@example.com",
                new BigDecimal("500.00"),
                "CREDIT_CARD",
                "EDUCATION"
        );
    }

    // ────────────────────────────────────────────────────────────────────────────────
    // PROCESS DONATION TESTS
    // ────────────────────────────────────────────────────────────────────────────────

    @Test
    @DisplayName("Should process valid donation and set COMPLETED status")
    void testProcessDonation_Success() {
        when(donationRepository.save(any(Donation.class))).thenAnswer(inv -> inv.getArgument(0));

        Donation result = donationService.processDonation(validDonation);

        assertNotNull(result);
        assertEquals(Donation.DonationStatus.COMPLETED, result.getStatus());
        assertNotNull(result.getTransactionId());
        assertTrue(result.getTransactionId().startsWith("TXN-"));
```

```
    assertNotNull(result.getDonationDate());
    verify(donationRepository, times(1)).save(any(Donation.class));
}

@Test
@DisplayName("Should set FAILED status when payment gateway declines (amount ends in .99)")
void testProcessDonation_PaymentFailed() {
    validDonation.setAmount(new BigDecimal("100.99"));
    when(donationRepository.save(any(Donation.class))).thenAnswer(inv -> inv.getArgument(0));

    Donation result = donationService.processDonation(validDonation);

    assertEquals(Donation.DonationStatus.FAILED, result.getStatus());
    assertTrue(result.getRemarks().contains("declined"));
}

@Test
@DisplayName("Should throw exception for null donation")
void testProcessDonation_NullDonation() {
    IllegalArgumentException ex = assertThrows(
            IllegalArgumentException.class,
            () -> donationService.processDonation(null)
    );
    assertEquals("Donation cannot be null.", ex.getMessage());
}

// ═══════════════════════════════════════════════════════════════════════════
// VALIDATION TESTS
// ═══════════════════════════════════════════════════════════════════════════

@Test
@DisplayName("Should throw exception for blank donor name")
void testValidation_BlankDonorName() {
    validDonation.setDonorName("");
    assertThrows(IllegalArgumentException.class,
            () -> donationService.validateDonation(validDonation));
```

```
}

@Test
@DisplayName("Should throw exception for invalid email format")
void testValidation_InvalidEmail() {
    validDonation.setDonorEmail("not-an-email");
    IllegalArgumentException ex = assertThrows(
        IllegalArgumentException.class,
        () -> donationService.validateDonation(validDonation)
    );
    assertTrue(ex.getMessage().contains("email"));
}

@Test
@DisplayName("Should throw exception when amount is below minimum ($1.00)")
void testValidation_AmountBelowMinimum() {
    validDonation.setAmount(new BigDecimal("0.50"));
    IllegalArgumentException ex = assertThrows(
        IllegalArgumentException.class,
        () -> donationService.validateDonation(validDonation)
    );
    assertTrue(ex.getMessage().contains("Minimum"));
}

@Test
@DisplayName("Should throw exception when amount exceeds maximum ($100,000)")
void testValidation_AmountAboveMaximum() {
    validDonation.setAmount(new BigDecimal("150000.00"));
    IllegalArgumentException ex = assertThrows(
        IllegalArgumentException.class,
        () -> donationService.validateDonation(validDonation)
    );
    assertTrue(ex.getMessage().contains("Maximum"));
}

@Test
```

```
@DisplayName("Should throw exception for invalid payment method")
void testValidation_InvalidPaymentMethod() {
    validDonation.setPaymentMethod("BITCOIN");
    IllegalArgumentException ex = assertThrows(
            IllegalArgumentException.class,
            () -> donationService.validateDonation(validDonation)
    );
    assertTrue(ex.getMessage().contains("Invalid payment method"));
}

@Test
@DisplayName("Should accept all valid payment methods")
void testValidation_AllValidPaymentMethods() {
    List<String> validMethods = List.of("CREDIT_CARD", "DEBIT_CARD", "UPI", "NET_BANKING");
    when(donationRepository.save(any())).thenAnswer(inv -> inv.getArgument(0));

    for (String method : validMethods) {
        validDonation.setPaymentMethod(method);
        assertDoesNotThrow(() -> donationService.processDonation(validDonation),
                "Should accept payment method: " + method);
    }
}

@Test
@DisplayName("Should throw exception for blank cause")
void testValidation_BlankCause() {
    validDonation.setCause("");
    assertThrows(IllegalArgumentException.class,
            () -> donationService.validateDonation(validDonation));
}

// ═══════════════════════════════════════════════════════════════════════════
// RETRIEVAL TESTS
// ═══════════════════════════════════════════════════════════════════════════

@Test
```

```
@DisplayName("Should return donation by ID")
void testGetDonationById_Found() {
    validDonation.setId(1L);
    when(donationRepository.findById(1L)).thenReturn(Optional.of(validDonation));

    Optional<Donation> result = donationService.getDonationById(1L);

    assertTrue(result.isPresent());
    assertEquals(1L, result.get().getId());
}

@Test
@DisplayName("Should return empty Optional for non-existent ID")
void testGetDonationById_NotFound() {
    when(donationRepository.findById(999L)).thenReturn(Optional.empty());

    Optional<Donation> result = donationService.getDonationById(999L);

    assertFalse(result.isPresent());
}

@Test
@DisplayName("Should return all donations for a valid email")
void testGetDonationsByEmail_Success() {
    List<Donation> donations = Arrays.asList(validDonation, validDonation);
    when(donationRepository.findByDonorEmail("john.doe@example.com")).thenReturn(donations);

    List<Donation> result = donationService.getDonationsByEmail("john.doe@example.com");

    assertEquals(2, result.size());
}

@Test
@DisplayName("Should throw exception for blank email in getDonationsByEmail")
void testGetDonationsByEmail_BlankEmail() {
    assertThrows(IllegalArgumentException.class,
```

```
        () -> donationService.getDonationsByEmail(""));
}

@Test
@DisplayName("Should return total of completed donations")
void testGetTotalDonations() {
    when(donationRepository.getTotalCompletedDonations()).thenReturn(new BigDecimal("1500.00"));

    BigDecimal total = donationService.getTotalDonations();

    assertEquals(new BigDecimal("1500.00"), total);
}

@Test
@DisplayName("Should return ZERO when no completed donations")
void testGetTotalDonations_NullReturnsZero() {
    when(donationRepository.getTotalCompletedDonations()).thenReturn(null);

    BigDecimal total = donationService.getTotalDonations();

    assertEquals(BigDecimal.ZERO, total);
}

// ═══════════════════════════════════════════════════════════════════════
// REFUND TESTS
// ═══════════════════════════════════════════════════════════════════════

@Test
@DisplayName("Should refund a COMPLETED donation")
void testRefundDonation_Success() {
    validDonation.setId(1L);
    validDonation.setStatus(Donation.DonationStatus.COMPLETED);
    when(donationRepository.findById(1L)).thenReturn(Optional.of(validDonation));
    when(donationRepository.save(any())).thenAnswer(inv -> inv.getArgument(0));

    Donation refunded = donationService.refundDonation(1L);
```

```
    assertEquals(Donation.DonationStatus.REFUNDED, refunded.getStatus());
    assertTrue(refunded.getRemarks().contains("Refund processed"));
}

@Test
@DisplayName("Should throw exception when refunding a non-COMPLETED donation")
void testRefundDonation_NotCompleted() {
    validDonation.setId(1L);
    validDonation.setStatus(Donation.DonationStatus.PENDING);
    when(donationRepository.findById(1L)).thenReturn(Optional.of(validDonation));

    assertThrows(IllegalStateException.class,
            () -> donationService.refundDonation(1L));
}

@Test
@DisplayName("Should throw exception when refunding non-existent donation")
void testRefundDonation_NotFound() {
    when(donationRepository.findById(999L)).thenReturn(Optional.empty());

    assertThrows(IllegalArgumentException.class,
            () -> donationService.refundDonation(999L));
}

// ═══════════════════════════════════════════════════════════════════════════
// BOUNDARY / EDGE CASE TESTS
// ═══════════════════════════════════════════════════════════════════════════

@Test
@DisplayName("Should accept minimum valid amount ($1.00)")
void testBoundary_MinimumAmount() {
    validDonation.setAmount(new BigDecimal("1.00"));
    when(donationRepository.save(any())).thenAnswer(inv -> inv.getArgument(0));

    assertDoesNotThrow(() -> donationService.processDonation(validDonation));
```

```
    }

    @Test
    @DisplayName("Should accept maximum valid amount ($100,000)")
    void testBoundary_MaximumAmount() {
        validDonation.setAmount(new BigDecimal("100000.00"));
        when(donationRepository.save(any())).thenAnswer(inv -> inv.getArgument(0));

        assertDoesNotThrow(() -> donationService.processDonation(validDonation));
    }

    @Test
    @DisplayName("Transaction ID should be unique per donation")
    void testTransactionId_Uniqueness() {
        when(donationRepository.save(any())).thenAnswer(inv -> inv.getArgument(0));

        Donation d1 = donationService.processDonation(new Donation("Alice", "alice@test.com",
                new BigDecimal("100"), "UPI", "HEALTH"));
        Donation d2 = donationService.processDonation(new Donation("Bob", "bob@test.com",
                new BigDecimal("200"), "UPI", "HEALTH"));

        assertNotEquals(d1.getTransactionId(), d2.getTransactionId());
    }
}
```

3.YAML File

Deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: donation-system
  namespace: production
  labels:
    app: donation-system
    version: "1.0.0"
spec:
  replicas: 3
```

```
selector:
  matchLabels:
    app: donation-system
strategy:
  type: RollingUpdate
  rollingUpdate:
    maxSurge: 1
    maxUnavailable: 0
template:
  metadata:
    labels:
      app: donation-system
  spec:
    containers:
      - name: donation-system
        image: yourdockerhubusername/online-donation-system:latest
        ports:
          - containerPort: 8080
        env:
          - name: SPRING_PROFILES_ACTIVE
            value: "production"
          - name: SPRING_DATASOURCE_URL
            valueFrom:
              secretKeyRef:
                name: donation-db-secret
                key: datasource-url
          - name: SPRING_DATASOURCE_USERNAME
            valueFrom:
              secretKeyRef:
                name: donation-db-secret
                key: datasource-username
          - name: SPRING_DATASOURCE_PASSWORD
            valueFrom:
              secretKeyRef:
                name: donation-db-secret
                key: datasource-password
```

```
        resources:
          requests:
            memory: "256Mi"
            cpu: "250m"
          limits:
            memory: "512Mi"
            cpu: "500m"
        readinessProbe:
          httpGet:
            path: /actuator/health
            port: 8080
          initialDelaySeconds: 30
          periodSeconds: 10
        livenessProbe:
          httpGet:
            path: /actuator/health
            port: 8080
          initialDelaySeconds: 60
          periodSeconds: 30
      restartPolicy: Always
```

Service.YAMl

```
apiVersion: v1
kind: Service
metadata:
  name: donation-system-service
  namespace: production
  labels:
    app: donation-system
spec:
  type: LoadBalancer
  selector:
    app: donation-system
  ports:
    - name: http
      protocol: TCP
```

```yaml
    port: 80
    targetPort: 8080

---
# ── Namespace for Staging ──────────────────────────────────────────
apiVersion: v1
kind: Namespace
metadata:
  name: staging

---
# ── Namespace for Production ───────────────────────────────────────
apiVersion: v1
kind: Namespace
metadata:
  name: production

---
# ── Secret template (fill with real values) ────────────────────────
apiVersion: v1
kind: Secret
metadata:
  name: donation-db-secret
  namespace: production
type: Opaque
stringData:
  datasource-url: "jdbc:postgresql://your-db-host:5432/donationdb"
  datasource-username: "donationuser"
  datasource-password: "your-secure-password"
```